Bahria University Lahore Campus

Department of Computer Science

Course Code/number	**GSC-122**
Course Title/Name	**Probability and Statistics**
Credit Hours/Contact hours	3
Degree Program	Bachelors of Science in Computer Science (BSCS)
Prerequisites or Co-requisites	None
Assessment Methods and Weightage	Quizzes 10 Assignments/Projects 20 Mid-Term Examination 20 Final Examination 50 **Total** **100**
Textbook (or **Laboratory Manual** for Laboratory Courses)	1. Ronald Walpole, Sharon L. Myers, Keying Ye "*Probability and Statistics for Engineers and Scientists*", 8th edition, 2008, Prentice Hall Publisher.
Reference Material (With Edition, ISBN#)	1. Richard l. Levin, David S. Rubin, Sanjay Rastogi, Masood Husain Siddique *"Statistics for Management"*, 7th edition, 2015, Prentice Hall Publisher
Web Resources/ URL (if any)	
Instructor Name/Cluster Head/Subject Expert	Instructor Name: Maham Kiran Designation: visiting Lecturer Status □ Visiting Cluster Head Name: Dr. Iram Noureen Subject Expert Name: Sumaira Noushen

Course Aims					
Course Objectives	1. To familiarize students with various statistical concepts and methods and enable them to develop statistical reasoning. 2. To understand and apply the concepts of Probability theory. 3. To provide students, the knowledge of different Probability distributions and their applications in Computer Science. 4. To enable the students to learn and apply the tools for curve fitting via Linear Regression and Correlation.				
Course Outcomes	After successful completion of the course, the students should be able to:			PLO	BT Level
	• **Explain** and **express** the basic understanding of probability and statistics.			1	C2
	• **Demonstrate** an ability to use descriptive techniques to describe the statistical data.			2	C2
	• **Apply** inferential statistical methods to solve problems.			3	C3
	• **Analyze** and investigate any given data distribution.			3	C4
Course Description/Catalogue	Introduction to Statistics and Data Analysis, Statistical Inference, Samples, Populations, and the Role of Probability. Sampling Procedures. Discrete and Continuous Data. Statistical Modeling. Types of Statistical Studies. Probability: Sample Space, Events, Counting Sample Points, Probability of an Event, Additive Rules, Conditional Probability, Independence, and the Product Rule, Bayes' Rule. Random Variables and Probability Distributions. Mathematical Expectation: Mean of a Random Variable, Variance and Covariance of Random Variables, Means and Variances of Linear Combinations of Random Variables, Chebyshev's Theorem. Discrete Probability Distributions. Continuous Probability Distributions. Fundamental Sampling Distributions and Data Descriptions: Random Sampling, Sampling Distributions, Sampling Distribution of Means and the Central Limit Theorem. Sampling Distribution of S2, t-Distribution, F-Quantile and Probability Plots. Single Sample & One- and TwoSample Estimation Problems. Single Sample & One- and Two-Sample Tests of Hypotheses. The Use of P-Values for Decision Making in Testing Hypotheses (Single Sample & One- and Two-Sample Tests), Linear Regression and Correlation. Least Squares and the Fitted Model, Multiple Linear Regression and Certain, Nonlinear Regression Models, Linear Regression Model Using Matrices, Properties of the Least Squares Estimators..				
	Week # and Date	Lecture/ Hour #	Topic to be covered	Learning outcomes	Reference Text
		1	Introduction to the course, Introduction to Statistics and its applications	Introduction to the course	Textbook and Lecture Handouts

		2	Fundamental concepts and definitions.	Students will learn the basic concepts that will be used in the course	Textbook and Lecture Handouts
		1	Descriptive Statistics and Inferential statistics, sample and population	Students should be able to understand the types of statistics	Textbook and Lecture Handouts
		2	frequency distribution, cumulative frequency, measurement scales	Students understanding of frequency distribution its representation and measurement scales	Textbook and Lecture Handouts
		1	Graphical Representation of Data, bar diagrams, histogram	An understanding graphical representation of data, bar diagram, histogram	Textbook and Lecture Handouts
		2	Frequency polygon, frequency curve, Stem and Leaf Plot, Dot Plot	Learn about frequency curve, polygon, dot plot, stem and leaf plot.	Textbook and Lecture Handouts
	4	1	Measure of central tendency, mean,	An understanding about mean, Geometric mean, harmonic mean	Textbook and Lecture Handouts
		2	Median, Mode	Learn about median and mod of data.	Textbook and Lecture Handouts
	5	1	Measure of dispersion, range	An understanding about measure of dispersion, range	Textbook and Lecture Handouts
		2	Quartile deviation, variance , standard deviation	Student will learn about quartile deviation, variance and standard deviation.	Textbook and Lecture Handouts
	6	1	Sample Space, Events, Counting Sample points	Learn different sample space, events, counting sample points.	Textbook and Lecture Handouts
		2	Probability of an event, Additive Rules and conditional probability	Learn different probability, additive rules and conditional probability.	Textbook and Lecture Handouts
	7	1	Product rules and Bayes' Rule, Independent and Dependent events	Understanding of product rules and Bayes'rule. Understanding of Independent and Dependent events.	Textbook and Lecture Handouts

		2	Random Variables, Types of random variables	A good understanding of Random Variables, Types of random variables.	Textbook and Lecture Handouts
	8	1	Probability distribution	Learn about Probability distribution.	Textbook and Lecture Handouts
		2	Mathematical Expectation, mean of a random variable	Learn about expectation, mean of a random variable.	Textbook and Lecture Handouts
	9		**Mid-Semester Exam**		
	10	1	Variance and covariance of random variable	An understanding of Variance and covariance of random variable.	Textbook and Lecture Handouts
		2	Means and Variances of Linear Combinations of Random Variables. Chebyshev's Theorem.	An understanding of Means and Variances of Linear Combinations of Random Variables. Learn about Chebyshev's Theorem.	Textbook and Lecture Handouts
	11	1	Discrete Probability Distributions, its mean, variance and standard deviation.	Able to understand Discrete Probability Distributions.	Textbook and Lecture Handouts
		2	Binomial Probability Distribution, its mean, variance and standard deviation.	Able to understand Discrete Binomial Probability Distributions.	Textbook and Lecture Handouts
	12	1	Continuous Probability Distributions. Normal Distribution.	Able to understand Continuous Probability Distributions.	Textbook and Lecture Handouts
		2	Application to Normal Distribution. Normal Approximation to Binomial Distribution.	Able to use the normal distribution in real life problems.	Textbook and Lecture Handouts
	13	1	Sampling distribution	Students will learn Fundamental Sampling	Textbook and Lecture Handouts

				Distributions and Data Descriptions.	
		2	Sampling Distribution of Means and the Central Limit Theorem	Learn Sampling Distribution of Means and the Central Limit Theorem.	Textbook and Lecture Handouts
	14	1	Sampling Distribution of S2	Learn Sampling Distribution of S2.	Textbook and Lecture Handouts
		2	t-Distribution	A good understanding of t-distribution.	Textbook and Lecture Handouts
	15	1	Linear Regression, Linear Regression Model Using Matrices, Scatter Plot.	Students will learn Linear Regression. An introduction to Linear Regression Model Using Matrices	Textbook and Lecture Handouts
		2	Least Squares and the Fitted Model	Students will learn Least Squares and the Fitted Model.	Textbook and Lecture Handouts
	16	1	Multiple Linear Regression	An introduction of Multiple Linear Regression.	Textbook and Lecture Handouts
		2	Nonlinear Regression Models, Correlation	Students will learn Nonlinear Regression Models. An introduction of Correlation.	Textbook and Lecture Handouts
	17	1	Properties of the Least Squares Estimators.	Students will learn Properties of the Least Squares Estimators.	Textbook and Lecture Handouts
		2	Revision	Revision	Textbook and Lecture Handouts
	18		**Final Term**		

Semester Calendar for Assignments/Quizzes/Project

Assignments/ Projects and quizzes Plan	Week #	Assignment No.	Quiz No.	Project Title	Assignment/Project Quiz Date	Result Date of Assignment/Project/Quiz
	1					
	2					

	3	Assignment-1				
	4		Quiz-1			
	5					Assignment-1, Quiz-1
	6	Assignment-2				
	7		Quiz-2			
	8					Assignment-2, Quiz-2
	10					
	11					
	12		Quiz-3			
	13	Assignment-3				
	14		Quiz-4			Assignment-3, Quiz-3
	15	Assignment-4				
	16					Assignment-4, Quiz-4
	17					

Instructor Name **Maham Kiran**

Instructor Signature _____

Date _____